

Adam Bouafia

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Education

Vrije Universiteit Amsterdam / Università degli Studi dell'Aquila, MSc (Double Degree) Sept 2023 – Dec 2025
in Computer Science

- First year at L'Aquila (2023-09 to 2024-06), second year at VU Amsterdam (2024-09 to 2025-12)
- Specialization: Software Engineering & Green IT
- Thesis: LogPress
 - Semantic Log Compression System

Higher Institute of Computer Science of Mahdia, BSc in Software Engineering & Information System Sept 2019 – June 2022

- Final Grade: 15.2/20
- Thesis: Flutter Mobile Security App 'M3ak'

Experience

Master's Thesis Researcher, Vrije Universiteit Amsterdam – Amsterdam, Netherlands Mar 2025 – Dec 2025

- LogPress Semantic Log Compression achieved 22.33x compression ratio with 100% lossless reconstruction across 2.4M+ logs
- Lazy Reconstruction query system with 21-111ms performance without full decompression

Researcher, Vivieb – Pescara, Italy Nov 2023 – Feb 2024

- Optimized Kubernetes cluster energy efficiency using Kubegreen and Grafana metrics

Software Developer, Euromed Organization – Paris, France July 2022 – Dec 2022

- Developed Dhayen mobile security app for Algerian national awareness campaign

Projects

Azure Landing Zone - Production MSP Infrastructure Feb 2026

- Production Azure infrastructure with hub-spoke topology, 100% Bicep, Azure Firewall & Bastion
- Multi-stage CI/CD pipeline (Azure DevOps) with approval gates and cost savings scripts

Greenmining - Python MSR Library for Green IT Jan 2026

- Python library on PyPI with 124 sustainability patterns from Green Software Foundation
- Multi-backend energy measurement (RAPL, CodeCarbon) with carbon footprint reporting

LCT - LLM Comparative Tool with Energy Profiling Oct 2025

- Open-source framework evaluating 500K+ HuggingFace LLMs across 17 algorithms with real-time energy profiling
- Dockerized CI/CD pipeline with carbon footprint analytics (tokens/joule, kg CO2)

CO2 Consumption Monitor (GNOME Shell Extension) Sept 2025

- GNOME extension on Gnome Store for real-time CO2 emissions tracking based on system usage
- Per-application tracking with ElectricityMaps integration and historical data export

Think Before You Save (Hackathon Winner - TU Delft)	Jan 2025
• Chrome extension calculating CO2 emissions from cloud storage, won Green Waves Hackathon	
Smart Fan and Gas Monitoring System	Sept 2024
• Real-time IoT monitoring using MQTT, Node-RED, InfluxDB, Grafana with Telegram alerts	

Certifications

JNCIA-DevOps (Juniper Networks Certified Associate - DevOps)	Nov 2024
IELTS Academic - English B2	May 2023

Skills

Programming Languages: Python, Java, JavaScript, C++ , Dart

Containerization and Orchestration: Kubernetes, Knative, Docker, Terraform

Serverless and Cloud Technologies: Serverless architecture, AWS Lambda, Knative

Benchmarking and Performance Evaluation: Experiment Runner, DeepEval, latency, throughput analysis

Data Monitoring and Visualization: InfluxDB, Grafana, Node-RED

Tools & Technologies: API Development, NLP, IoT, ML frameworks, CI/CD pipelines (Jenkins/GitLab CI)

Languages

English: Fluent

Arabic: Native

French: Proficient

Italian: Proficient

Dutch: Beginner